GnssProcessor: Mathematical Reference

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Contents

1 Introduction

This document is the mathematical companion to `GnssProcessor`, a C++17 post-processing and real-time GNSS engine implementing PPP, PPP-RTK, and double-differenced baseline modes. Its scope is deliberately narrow:

- State vector layout (**??**) — which states the filter carries and when each is enabled.
- Observation equations (**??**) — the *uncombined* pseudorange and carrier-phase models the filter linearizes around.
- Differencing transformations (**??**) — the between-satellite single-difference (BSSD) used in PPP-RTK and the double-difference (DD) used in baseline modes, including the S-transform structure that maps undifferenced parameters to differenced ones.

Implementation details (function signatures, class responsibilities, config field reference) live in the headers under `include/` and are rendered via Doxygen. When this document references a specific algorithmic component, it cites the source file directly (e.g. `src/core/filter/KalmanFilter.cpp`).

The notation follows the IGS conventions adopted by **subirana2013gnss** and **teunissen2017handbook**; see **??** for the symbol table this document depends on.

2 Notation and conventions

Indices and sets. We use lower-case Latin letters for indices and calligraphic upper-case for sets:

- $r \in \mathcal{R}$ — receiver. Single-receiver modes (PPP, PPP-RTK) have $|\mathcal{R}| = 1$; baseline modes have a fixed reference and one or more rovers.
- $g \in \mathcal{G} = \{\text{GPS}, \text{GAL}, \text{BDS}, \text{GLO}, \text{QZS}, \dots\}$ — constellation.
- $i \in \mathcal{S}_g$ — satellite of constellation g. p_g denotes the pivot satellite of constellation g under the BSSD parameterisation (**??**).
- $s \in \mathcal{B}$ — signal/observation type, identified by its RINEX-3 three-character code (e.g. `C1C`, `L2W`, `L5X`).
- $k \in \mathbb{N}$ — discrete epoch index.

Vectors and matrices. Bold lower-case for vectors ($\boldsymbol{x}$, $\boldsymbol{r}$); bold upper-case for matrices ($\boldsymbol{H}$, $\boldsymbol{P}$). $\boldsymbol{a}^{\mathrm{T}}$ is the transpose. A hat denotes a state estimate; a tilde a one-step prediction: $\hat{\boldsymbol{x}}_{k|k}$ vs. $\tilde{\boldsymbol{x}}_{k|k-1}$.

Observables. For each receiver/satellite/signal triple, the engine consumes two real-valued observables (in metres) per epoch:

- $P^i_{r,s}$ — pseudorange (code), in metres.
- $\Phi^i_{r,s}$ — carrier phase, scaled to metres via λ_s. The accumulated cycle count $\Phi^i_{r,s} \cdot \lambda_s^{-1}$ is the integer ambiguity plus a fractional bias.

Where the distinction matters, s may further denote the band ($L_1, L_2, L_5, \ldots$) and the tracking attribute (e.g. `C`, `W`, `X`).

Frequencies and wavelengths. f_s is the carrier frequency (Hz) and $\lambda_s = c/f_s$ the wavelength (m). The frequency-squared scaling factor used to project the L1-equivalent slant ionospheric delay onto signal s is

$$\mu_s \;:=\; \frac{f_1^2}{f_s^2}. \tag{1}$$

Differencing operators. Three differencing operations recur throughout the document:

$$\Delta q^{i,p_g}_{r,s} := q^i_{r,s} - q^{p_g}_{r,s}, \quad \nabla q^i_{r_1 r_2,s} := q^i_{r_1,s} - q^i_{r_2,s}, \quad \nabla\Delta q^{i,p_g}_{r_1 r_2,s} := \Delta\big(\nabla q^i_{r_1 r_2,s}\big).$$

Read Δq as "between-satellite single difference of q relative to pivot p_g," ∇q as "between-receiver single difference," and $\nabla\Delta q$ as "double difference."

Sign conventions. We adopt the convention that the slant ionospheric delay $I^i_{r,s}$ enters *additively* on the code observable and *subtractively* on the carrier phase observable; this is the source of the well-known "code–phase divergence." All other range-additive terms (geometry, clocks, troposphere, biases) carry the same sign on both observables. See **????** for the explicit equations.

Symbol table.

Symbol	Units	Meaning
$\boldsymbol{r}_r, \boldsymbol{r}^i$	m	ECEF position of receiver/satellite
ρ^i_r	m	geometric range $\lVert\boldsymbol{r}^i - \boldsymbol{r}_r\rVert$
$\boldsymbol{e}^i_r$	—	unit LOS from r to i in ECEF
$c\,\mathrm{d}t_r$	m	receiver clock offset (scaled by c)
$c\,\mathrm{d}t^i$	m	satellite clock offset (from CLK product)
$c\,\delta t^{(g)}_r$	m	receiver inter-system time offset for constellation $g \neq g_{\mathrm{ref}}$
T^i_r	m	total tropospheric path delay
$\mathrm{ZHD}^i_r, \mathrm{ZWD}^i_r$	m	zenith hydrostatic / wet components
$I^i_{r,1}$	m	L1-equivalent slant ionospheric delay
$N^i_{r,s}$	cyc	integer carrier-phase ambiguity
$b_{r,s}, b^i_s$	m	receiver / satellite phase biases
$B_{r,s}, B^i_s$	m	receiver / satellite code biases (OSB)
$\varepsilon^P, \varepsilon^\Phi$	m	residual (multipath + thermal)

3 State vector layout

The Kalman filter operates on a state vector $\boldsymbol{x}_k \in \mathbb{R}^{n_k}$ whose dimension n_k *varies across epochs*: per-satellite and per-(satellite, signal) substates are added when a new sat/signal is first observed and pruned (or marked stale) when the satellite drops below the elevation mask or its phase track is interrupted by a cycle slip without recovery. All bookkeeping — key→index maps, free-list re-use, and dimensional reshaping of $\boldsymbol{x}$ and $\boldsymbol{P}$ — is encapsulated in `StateManager`; `KalmanFilter` consumes the state through that interface and never indexes raw memory by hard-coded offsets.

3.1 Schematic layout

In schematic form, suppressing the epoch index k:

$$\begin{aligned}\boldsymbol{x} \;=\; \big[\; & \underbrace{\boldsymbol{r}_r^{\mathrm{T}}}_{3},\; \underbrace{\boldsymbol{v}_r^{\mathrm{T}}}_{0 \text{ or } 3},\; \underbrace{c\,\mathrm{d}t_r}_{1},\; \underbrace{(c\,\delta t_r^{(g)})_{g\neq g_{\mathrm{ref}}}^{\mathrm{T}}}_{|\mathcal{G}|-1},\\ & \underbrace{\mathrm{ZWD}_r,\,(g_N, g_E)}_{1+0 \text{ or } 1+2},\; \underbrace{(c\,\mathrm{d}t_r^{\phi})}_{0 \text{ or } 1},\\ & \underbrace{(I_{r,1}^i)_{i\in\mathcal{V}_k}^{\mathrm{T}}}_{|\mathcal{V}_k|},\; \underbrace{(N_{r,s}^i)_{(i,s)\in\mathcal{A}_k}^{\mathrm{T}}}_{|\mathcal{A}_k|},\\ & \underbrace{(B_{r,s})_{s\in\mathcal{B}}^{\mathrm{T}}}_{|\mathcal{B}|},\; \underbrace{(B_s^i)_{(i,s)\in\mathcal{D}_k}^{\mathrm{T}}}_{|\mathcal{D}_k|}\;\big]^{\mathrm{T}}. \end{aligned} \tag{2}$$

where

- $\mathcal{V}_k$ is the set of satellites currently visible at epoch k (i.e. above the elevation mask, with valid orbit/clock products and at least one tracked signal),
- $\mathcal{A}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the set of phase-tracked (satellite, signal) pairs,
- $\mathcal{D}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the subset of (satellite, signal) pairs for which the BIA product lacks a satellite OSB and a state-space estimate is therefore created (see `estimation.estimate_sat_biases`).

3.2 Component-by-component description

Position $\boldsymbol{r}_r \in \mathbb{R}^3$. ECEF Cartesian receiver position. Always present. Initialised either from the RINEX header approximate position or, after the first epoch, from a Weighted Least-Squares (WLS) seed solution to within the linearisation radius of the geometric range model (**??**).

Velocity $\boldsymbol{v}_r \in \mathbb{R}^3$. Optional. Enabled by `estimation.estimate_velocity = true`, or forced on by tightly-coupled IMU integration. When present, the position-velocity block evolves as a discrete-time integrated random walk under the kinematic process model with spectral densities `position_q` and `velocity_q` respectively.

Receiver clock $c\,\mathrm{d}t_r$. Single state, in metres. Process noise is large (`clock_bias_q` $\sim 10^3\,\mathrm{m}^2$ per second) so the state is effectively re-estimated each epoch; this captures oscillator drift without imposing a stochastic model that may not match the receiver's actual oscillator.

Inter-system time offsets $c\,\delta t_r^{(g)}$. One per active constellation other than the chosen reference (default GPS). These absorb the per-constellation clock biases that arise from mixing analysis-centre-defined clock products across systems ("GAL − GPS time offset," etc.).

Tropospheric ZWD and gradients. ZWD_r models the zenith wet delay; the zenith hydrostatic component ZHD_r is treated deterministically via the Saastamoinen model [**saastamoinen1972atmospheric**]. Total slant tropospheric delay is then

$$T_r^i \;=\; m_h(\theta_r^i)\,\mathrm{ZHD}_r \;+\; m_w(\theta_r^i)\,\mathrm{ZWD}_r \;+\; m_w(\theta_r^i)\cot\theta_r^i\big[g_N\cos\alpha_r^i + g_E\sin\alpha_r^i\big], \tag{3}$$

with elevation θ_r^i, azimuth α_r^i, and the Niell or VMF mapping functions m_h, m_w. Gradients (g_N, g_E) are optional.

Per-satellite slant ionosphere $I_{r,1}^i$. PPP-RTK-only. One state per visible satellite, expressed at the L1 frequency; signal-specific delays follow **??**. Initial value seeded from the Klobuchar broadcast model when no external ionospheric product is available. Modelling these per-sat states explicitly is what permits PPP-RTK to retain integer ambiguity recovery without forming the ionosphere-free combination (which amplifies code noise by a factor of ~ 3).

Carrier-phase ambiguities $N_{r,s}^i$. One state per phase-tracked (i, s) pair, in cycles. Treated as float during the prediction/update cycle and resolved to integers post-hoc by the LAMBDA-family algorithms in `filter/AmbiguityResolver` [**teunissen1995lambda**] when AR is enabled. Cycle slips trigger a re-seed via `StateManager::resetAmbiguity`, which inflates the diagonal covariance entry while preserving the cross-state correlations that permit a consistent re-fix.

Receiver code biases $B_{r,s}$. One per signal in $\mathcal{B}$. These absorb the receiver-side inter-frequency biases that the OSB convention does not reference to zero.

Satellite code biases B_s^i. Per (satellite, signal) pair, but *only when the BIA file does not supply an OSB* for that pair — typically a third-frequency signal in a triple-frequency setup. When present, these states are weakly stochastic (`sat_bias_q` $\sim 10^{-10}$) since the underlying physical biases are nearly stationary on the daily timescale.

3.3 Dynamic indexing

The variable-dimension blocks (I, N, B^i) are addressed via key maps maintained by `StateManager`. The canonical key formats are defined once in `include/filter/StateKeySchema.hpp`:

- iono key: `"<sys><prn>"`, e.g. `"G05"` for GPS PRN 5;
- ambiguity key: `"<sys><prn>_<signal>"`, e.g. `"G05_L1C"`;
- satellite-bias key: `"<sys><prn>_<signal>"`, sharing the ambiguity-key namespace but stored under a distinct map.

This single source of truth eliminates the off-by-one errors that plague hand-managed dynamic state vectors and makes the BSSD/DD S-transforms (**??**) implementable as block operations on the relevant key sets.

4 Observation equations

This section gives the full *undifferenced* observation model that the filter linearises around at each epoch. The PPP-RTK parameterisation is uncombined: per-signal pseudorange and carrier phase are consumed directly, without forming ionosphere-free or wide-lane combinations as a preprocessing step. Combinations (IF, MW, GF) are still computed for diagnostic and slip-detection purposes (see `slip/MwSlipDetector`, `slip/GfSlipDetector`) but they do not enter the measurement update path.

4.1 Code (pseudorange) observation

For receiver r tracking satellite i on signal s, the observed pseudorange is

$$\boxed{P^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;+\; \mu_s\, I^i_{r,1} \;+\; B_{r,s} - B^i_s \;+\; \varepsilon^P_{r,i,s}.} \tag{4}$$

4.2 Carrier-phase observation

The corresponding carrier-phase observation, scaled to metres by λ_s, is

$$\boxed{\Phi^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;-\; \mu_s\, I^i_{r,1} \;+\; \lambda_s\, N^i_{r,s} \;+\; b_{r,s} - b^i_s \;+\; \varepsilon^\Phi_{r,i,s}.} \tag{5}$$

The defining structural difference between **????** is the sign of the ionospheric term: $+\mu_s I$ on code, $-\mu_s I$ on phase. Three additional remarks:

1. The phase ambiguity $N^i_{r,s}$ is integer in principle but is contaminated by the receiver-plus-satellite phase biases $b_{r,s} - b^i_s$. In a *float* solution these biases are absorbed into $N^i_{r,s}$, leaving a non-integer state with no structural advantage over the code observation. Recovering integer-valued ambiguities therefore requires either external Fractional Cycle Bias (FCB) products or differencing (**??**), which cancels common-mode bias terms.
2. The clock product $c\,\mathrm{d}t^i$ is read from the analysis-centre CLK file and treated as deterministic in the filter; any error in the product appears as residual on the observation and is absorbed by the receiver clock state.
3. The system time offset $c\,\delta t_r^{(g(i))}$ is zero by construction when $g(i)$ is the reference constellation (default GPS). Together with the receiver clock $c\,\mathrm{d}t_r$ this means a multi-constellation run carries $|\mathcal{G}|$ clock-like states in total.

4.3 Geometric range and linearisation

The geometric range $\rho^i_r = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|_2$ enters **????** non-linearly through the receiver position $\boldsymbol{r}_r$. The filter linearises about the current state estimate $\hat{\boldsymbol{r}}_r$:

$$\rho^i_r \;=\; \hat{\rho}^i_r \;-\; (\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}(\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r) \;+\; \mathcal{O}\big(\|\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r\|^2\big)\,, \tag{6}$$

with the unit line-of-sight

$$\boldsymbol{e}^i_{\hat{r}} \;:=\; \frac{\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r}{\|\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r\|_2}. \tag{7}$$

The Jacobian row $\boldsymbol{H}_{(i,s)}$ contributing to the design matrix at the position columns is therefore $-(\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}$. For sub-metre residual targets this single linearisation per epoch is sufficient; the engine

optionally relinearises within the epoch (`processing.relinearize_geometry`, on by default) to handle long initial position offsets.

The satellite position $\boldsymbol{r}^i$ is itself a function of the transmission time, which depends on ρ_r^i/c and therefore on $\boldsymbol{r}_r$. Resolving this implicit dependency by fixed-point iteration (typically 1–2 sweeps) is performed by `model/ModelGeometry`.

4.4 Stochastic model

Observation noise covariance is structured per epoch as a block diagonal matrix with entries

$$\sigma^2_{r,i,s} \;=\; \sigma^2_{0,\mathrm{type}(s)} \;\cdot\; w(\theta_r^i)^{-1} \;\cdot\; \xi(\mathrm{C/N_0 \text{ at } } (r,i,s)), \tag{8}$$

combining a per-type baseline ($\sigma_{0,\mathrm{code}}$, $\sigma_{0,\mathrm{phase}}$ from `filter.measurement_noise`), an elevation-dependent down-weighting factor $w(\theta) \in \{1, \sin\theta, \sin^2\theta, \dots\}$ selected by `filter.stochastic_model`, and an optional carrier-to-noise ratio scaling ξ parameterised by `snr_model_params`. Cross-correlations between observations are assumed zero in the undifferenced parameterisation; differencing introduces structured correlations handled in **??**.

5 Differencing transformations

PPP-RTK and baseline modes do not consume the undifferenced observations of **????** directly. Each forms a linear combination across satellites (BSSD), receivers (BRSD), or both (DD), absorbing common-mode terms into the cancellation and recovering integer-friendly residual structure. This section gives both the algebraic cancellations and the matrix-form S-transforms that map undifferenced parameters to differenced ones.

5.1 Between-satellite single difference (BSSD)

For each constellation g, choose a pivot satellite p_g (typically the highest-elevation tracked sat with a stable phase lock; see `filter/PivotManager`, "sticky pivot" policy). For any non-pivot satellite $i \in \mathcal{S}_g \setminus \{p_g\}$ and signal s, define

$$\begin{aligned} \Delta P^{i,p_g}_{r,s} &:= P^i_{r,s} - P^{p_g}_{r,s}, \\ \Delta \Phi^{i,p_g}_{r,s} &:= \Phi^i_{r,s} - \Phi^{p_g}_{r,s}. \end{aligned} \tag{9}$$

Substituting **????** and cancelling the receiver-side common terms ($c\,\mathrm{d}t_r$, $c\,\delta t_r^{(g)}$, $B_{r,s}$, $b_{r,s}$) yields

$$\boxed{\Delta P^{i,p_g}_{r,s} \;=\; \Delta\rho^{i,p_g}_r - c\,\Delta \mathrm{d}t^{i,p_g} + \Delta T^{i,p_g}_r + \mu_s\,\Delta I^{i,p_g}_{r,1} - \Delta B^{i,p_g}_s + \Delta\varepsilon^{P},} \tag{10}$$

$$\boxed{\Delta \Phi^{i,p_g}_{r,s} \;=\; \Delta\rho^{i,p_g}_r - c\,\Delta \mathrm{d}t^{i,p_g} + \Delta T^{i,p_g}_r - \mu_s\,\Delta I^{i,p_g}_{r,1} + \lambda_s\,\Delta N^{i,p_g}_{r,s} - \Delta b^{i,p_g}_s + \Delta\varepsilon^{\Phi}.} \tag{11}$$

The cancellations are exact only when both observations enter the filter with the same modelling: same tropospheric mapping (which is why ΔT is small but not zero), same antenna PCV correction, and so on.

Pivot transitions. When the incumbent pivot leaves the visible set, p_g is re-elected from the remaining satellites. The float ambiguities $\{\Delta N^{i,p_g^{\mathrm{old}}}_{r,s}\}_{i,s}$ must be re-expressed relative to p_g^{new}. Rather than discarding and re-seeding (which destroys the integer information accumulated since

the last fix), the engine applies the linear S-transformation $\boldsymbol{x}^{\text{new}} = \boldsymbol{S}\,\boldsymbol{x}^{\text{old}}$ with $\boldsymbol{S}$ chosen so that $\Delta N_{r,s}^{p_g^{\text{new}},p_g^{\text{old}}}$ becomes the new reference (zero). The covariance transforms as $\boldsymbol{P}^{\text{new}} = \boldsymbol{S}\,\boldsymbol{P}^{\text{old}}\,\boldsymbol{S}^{\mathrm{T}}$; see `filter/PivotManager` for the implementation and `baseline.enable_pivot_handover` for the governing config flag.

5.2 Between-receiver single difference (BRSD)

In baseline modes there is at least one rover r_{rov} and exactly one fixed-coordinate reference r_{ref}. The BRSD operator forms

$$\nabla P^i_{r_{\text{rov}}r_{\text{ref}},s} \;:=\; P^i_{r_{\text{rov}},s} - P^i_{r_{\text{ref}},s}, \tag{12}$$

and analogously for $\nabla\Phi$. The cancellations produced by $\nabla\cdot$ are dual to those produced by $\Delta\cdot$: BRSD cancels *satellite*-side common terms ($c\,\mathrm{d}t^i$, B^i_s, b^i_s) but preserves receiver clock and biases of *both* stations. No state-space estimation of satellite biases is required when ∇ is applied — they disappear before reaching the filter.

5.3 Double difference (DD)

The composition $\nabla\Delta\cdot = \Delta\big(\nabla\,\cdot\big)$ is order-independent up to noise correlation: $\Delta\big(\nabla\,\cdot\big) = \nabla\big(\Delta\,\cdot\big)$. Applying it to the carrier-phase model and collecting common-mode cancellations:

$$\nabla\Delta\Phi^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},s} \;=\; \nabla\Delta\rho^{i,p_g}_{r_{\text{rov}}r_{\text{ref}}} + \nabla\Delta T^{i,p_g}_{r_{\text{rov}}r_{\text{ref}}} - \mu_s\,\nabla\Delta I^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},1} + \lambda_s\,\nabla\Delta N^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},s} + \nabla\Delta\varepsilon^{\Phi}. \tag{13}$$

What survives in **??** is precisely the geometry term (linear in the rover position perturbation), the differential atmospheric delays, and the integer DD ambiguity — the structural form that makes carrier-phase RTK possible.

Short vs. long baselines. On short baselines ($\lesssim 10\,\text{km}$) the differential atmospheric terms in **??** are negligible at the millimetre level and need not be modelled. As baseline length grows, the differential ionosphere $\nabla\Delta I$ scales linearly with separation and saturates at typically 1 m for separations of a few hundred kilometres, swamping the carrier wavelength. Two mitigations are available in `GnssProcessor`:

- `baseline.use_iono_free` forms the dual-frequency ionosphere-free combination at the DD level, eliminating the first-order ionospheric term at the cost of a non-integer ambiguity (AR is automatically excluded for IF rows) and $\sim 3\times$ amplified code noise;
- `baseline.estimate_differential_tropo` adds a single $\Delta\text{ZWD}_{r_{\text{rov}}r_{\text{ref}}}$ state with slow random-walk dynamics, capturing the residual differential wet delay that survives the BRSD on baselines with significant altitude difference.

5.4 S-transform structure

Both BSSD and DD admit a matrix-form representation as $\boldsymbol{x}_{\text{diff}} = \boldsymbol{S}\,\boldsymbol{x}_{\text{undiff}}$ with $\boldsymbol{S}$ acting as a block design transform. For BSSD restricted to the ambiguity block of one constellation g on one signal s, with $|\mathcal{S}_g|$ tracked sats, $\boldsymbol{S}$ is a $(|\mathcal{S}_g|-1)\times|\mathcal{S}_g|$ matrix of the form

$$\boldsymbol{S}_{\text{BSSD}} \;=\; \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & & & \\ 0 & 0 & \cdots & 0 & -1 & 0 & \cdots & 1 \end{bmatrix}, \tag{14}$$

where the column of -1's sits at the index of the pivot p_g. The DD case stacks BSSD on top of an analogous between-receiver $\boldsymbol{S}_{\mathrm{BRSD}}$, yielding $\boldsymbol{S}_{\mathrm{DD}} = \boldsymbol{S}_{\mathrm{BSSD}} \otimes \boldsymbol{S}_{\mathrm{BRSD}}$ when the satellite and receiver dimensions are factorisable. The transformed covariance is $\boldsymbol{P}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{P}_{\mathrm{undiff}}\,\boldsymbol{S}^{\mathrm{T}}$, a generally dense matrix even when $\boldsymbol{P}_{\mathrm{undiff}}$ is diagonal (because each differenced row borrows from the same pivot column). This dense correlation structure is the algebraic reason DD-based filters cannot be naively implemented as independent per-pair scalar updates — the correctly correlated joint update is what unlocks integer ambiguity recovery via the LAMBDA decorrelation. The general theory and a thorough catalogue of admissible S-transforms is given in **teunissen1985generalised** and in the rank-defect treatment of **deJonge1998processing**.